

Scalable Manufacturing of Low-Symmetry Plasmonic Nanospindle Arrays with Tunable Surface Lattice Resonance

Hongyan Li, Jingyi Zhao, Yazhi Wang, Haitao Liu, Qianyun Chen, Yilin Bao, Miaoen Zhou, Yue Li, Yutao Sang, Fan Yang,* and Zhihong Nie*



Cite This: *ACS Nano* 2025, 19, 7391–7400



Read Online

ACCESS |

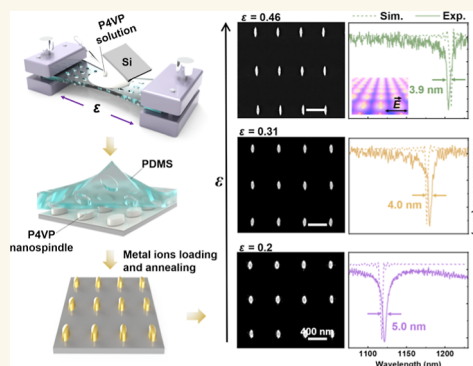
Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: Geometry-dependent plasmonic surface lattice resonances (SLRs) have garnered great interest across a range of applications, including nanolasers, sensors, photocatalysis, and nonlinear optics. However, the rational fabrication of high-quality, low-symmetry, plasmonic nanoparticle arrays over large areas remains challenging. Herein, we report a versatile strategy for the scalable fabrication of centimeter-scale plasmonic nanospindle (NS) arrays with high positional and orientational precision. Our approach combines solvent-assisted soft lithography with in situ reduction of metal precursors, enabling the cost-effective production of large-area and well-ordered NS arrays without the need of specialized equipment. The Au NS arrays exhibit superior SLRs with a ultranarrow line width of 3.9 nm and a quality factor (*Q*-factor) of 309. The aspect ratio and lattice geometry of the NSs can be precisely tuned by applying mechanical strain to the stretchable elastomeric template, thus, allowing us to customize the SLR performance across the near-infrared spectrum. This technique enables the precise engineering of anisotropic nanoparticle arrays in a standard chemistry laboratory, opening new possibilities for advanced plasmonic devices.

KEYWORDS: stretchable, anisotropic nanoparticles, array, scalable, plasmonic, surface lattice resonance



Plasmonic metal nanoparticles (NPs) are of broad interest owing to their strong interaction with light and their efficient light confinement at subwavelength scales. This light–NP interaction induces the collective oscillation of free electrons commonly known as localized surface plasmon resonance (LSPR).^{1,2} The localized excitations accompanied by large near-field enhancement have enabled enormous applications of plasmonic NPs in biosensing,^{3–5} photocatalysis,⁶ nanophotonics,^{7,8} and nonlinear optics.^{9,10} However, due to the intrinsic Ohmic losses present in metals at optical frequencies, LSPR-based plasmonic resonances always have a wide line width and a low quality-factor (*Q*-factor),^{11–13} which limit the potential applications of plasmonic-based devices, such as nanolasing^{14,15} and sensing.¹⁶ One effective approach toward obtaining a high-*Q* resonance is arranging plasmonic nanostructures periodically to achieve a hybrid excitation called surface lattice resonances (SLRs). The SLR modes are LSPR mode of individual NP coupling to the in-plane diffraction orders of the periodic array, which supports an ultranarrow plasmon resonance toward a theoretical limit of ~1–2 nm.^{13,15,17} However, such high-*Q* resonance has been rarely achieved in experiments, largely due to the difficulty in fabrication of nanostructures with a complex geometry, limited

throughput for large-area fabrication, and poor spatial coherence of light source.^{13,18–20}

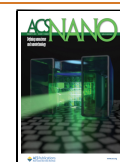
Recent theoretical studies have predicted that the nanostructure geometry dictates its coupling efficiency to free space.²¹ Experiments demonstrate that the coupling efficiency of anisotropic structures such as nanorods is much higher than that of isotropic structures, enabling the achievement of high-*Q* resonance.²¹ Additionally, anisotropic metallic NPs can exhibit polarization-dependent optical properties that cannot be readily obtained by spherical NPs.^{20,22–24} So far, the top-down technique (e.g., e-beam lithography) is the main strategy for fabricating high-quality arrays of anisotropic gold NPs. This approach, however, involves costly and time-consuming processes, which makes it difficult to achieve large areas.^{20,21,25–27} Another existing fabrication strategy involves

Received: December 19, 2024

Revised: February 2, 2025

Accepted: February 4, 2025

Published: February 14, 2025



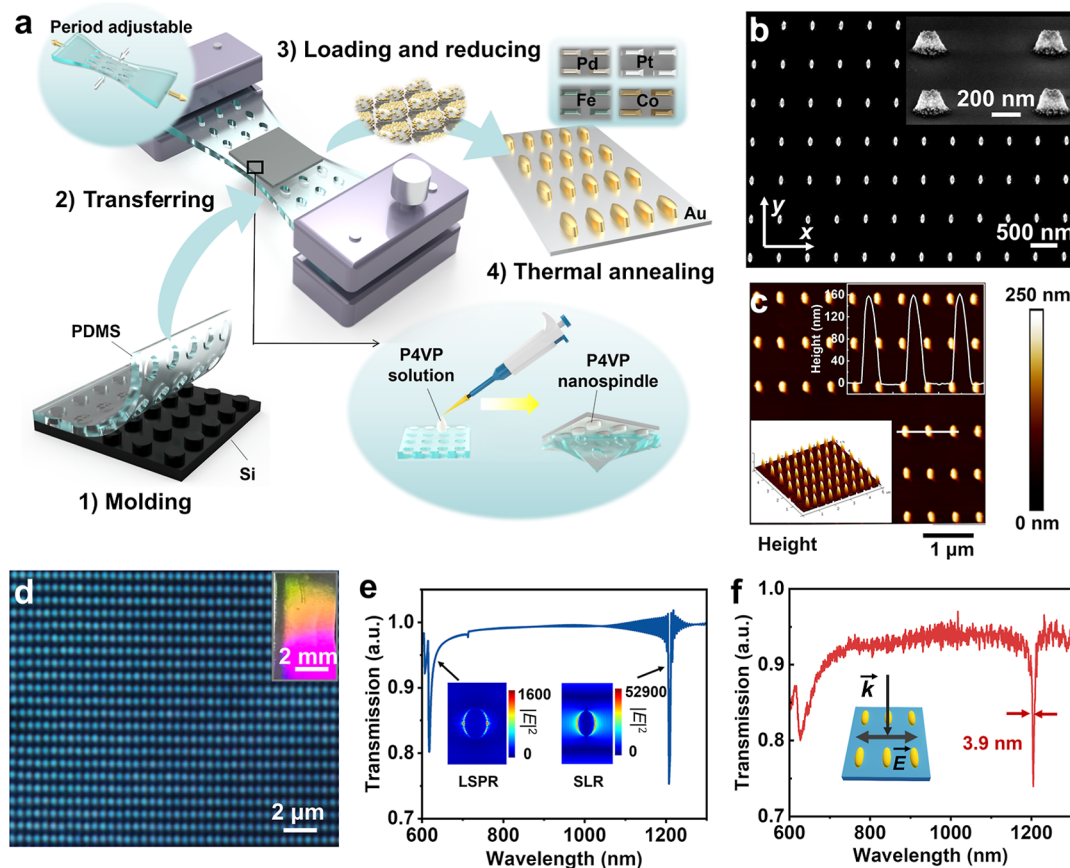


Figure 1. Fabrication of inorganic NS array with tunable lattice parameters. (a) Schematic illustration of the nanofabrication process. (b) SEM images of gold NS array. Inset is the cross-sectional SEM image of NS array at a tilted angle of 45° along the longitudinal axis. (c) 2D and 3D (inset) atomic force microscopy images of GNS array in a tapping mode. The upper inset shows the height profiles of the GNS. (d) Dark-field microscopy image of GNS array. Inset is an optical micrograph of the corresponding large-area GNS array. (e,f) Simulated (e) and measured (f) transmission spectrum of the GNS array under x -polarized light, surrounded by a dimethyl sulfoxide ($n = 1.48$) environment, which is used to match the refractive index of the substrate. Insets in (e) show the simulated magnitude of the electric field $|E|^2$ for the entire unit cell of both LSPR and SLR modes in the x - y plane and inset in (f) illustrates the direction of incident electric field.

the placement of gold nanorods into lithographically fabricated hole-like patterns with a defined shape^{28,29} through electric fields,^{28,30} magnetic field interactions,²⁹ or capillary forces.³¹ However, it remains challenging to control the accurate position and orientation of NPs, and hence, the lattice parameters (e.g., periodicity, symmetry, size). Nevertheless, there is an urgent need for new approaches to efficiently fabricate large-scale anisotropic NP arrays in a high-precision and flexible manner.

Herein, we present a versatile strategy for the large-area ($7 \times 5 \text{ mm}^2$) fabrication of inorganic nanospindle (NS) array with high position precision and uniform orientation. This strategy mainly involves patterning polymeric nanoarrays with targeted morphologies, chelating metal precursors, in situ reduction of metals, and thermal annealing of NP arrays (Figure 1a). The flexible nature of elastomeric templates provides a convenient way to manipulate the morphology of NP units. By applying different strains on the templates, the aspect ratio of NS can be precisely tuned in the range of 2.1 to 3.4, and the periodic spacing along the x axis (P_x) and y axis (P_y) can be varied in the range of 555 ± 10 – 480 ± 11 and 750 ± 11 – 840 ± 12 nm, respectively. Owing to the high degree of order in the arrangement of NSs with minimal defects, the Au NS array displayed an ultranarrow SLR resonance with a line width of ~ 3.9 nm and a Q -factor of ~ 309 . This Q -factor is the highest

value of gold arrays reported in the literature when a tungsten-halogen lamp with a relatively poor coherence is used as a light source.^{15,21,32,33} The SLR peaks can be tuned in the near-infrared range from 1120 to 1205 nm by varying the strains applied to the polymeric nanoarrays. Moreover, strong electric field enhancement ($|E/E_0|^2 = 53000$) was supported at the resonance of SLR. This versatile fabrication technique holds the promise for advancing the field of biosensors,³⁴ photonic devices,³⁵ photoelectrocatalysis,³⁶ etc.

RESULTS AND DISCUSSION

Fabrication of NS Arrays with Tunable Lattice Parameters. The strategy we developed involves the soft lithographic fabrication of polymeric nanocylindroid arrays using elastomeric stamps in a stretched state, loading of inorganic salt precursors into polymeric nanodomains, reducing inorganic salts by oxygen plasma, and thermal annealing of the NS structures (Figure 1a and see more details in the methods). In a typical system, a relief silicon master with nanopillars (diameter $d = 190$ nm, height $h = 350$ nm, and period $p_0 = 600$ nm) was molded against elastomeric poly(dimethylsiloxane) (PDMS) to produce soft stamps patterned with circular holes (step 1). Because of a Poisson's ratio (which is the negative of the ratio of transverse to axial strain) of 0.5 for PDMS, stretching in the x -direction results in

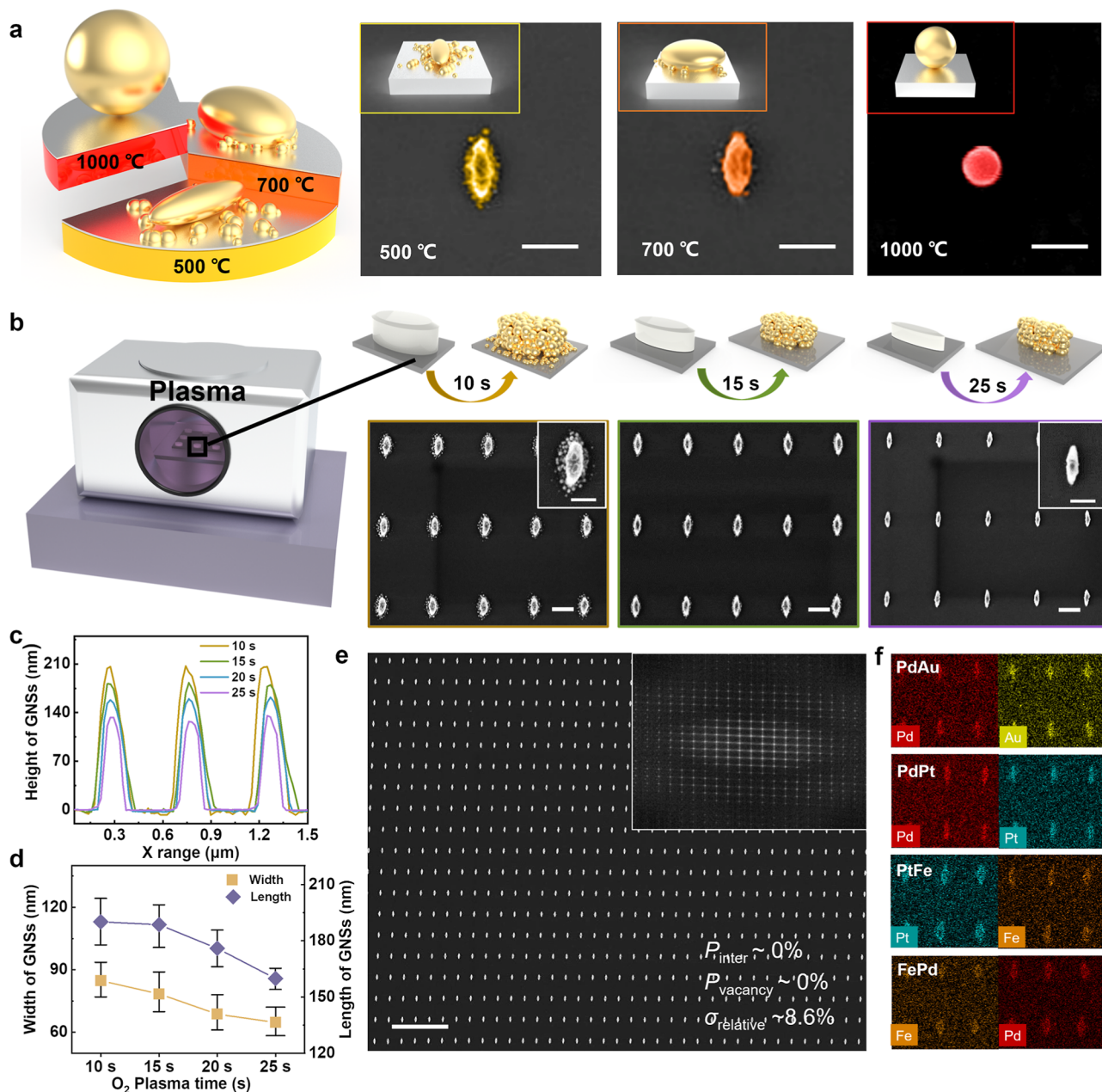


Figure 2. Effect of thermal annealing processes and O₂ plasma treatment on the structure of GNS array. (a) SEM images of GNSs obtained at thermal annealing temperatures of 500, 700, and 1000 °C, with corresponding schematic illustrations in the left image and insets. (b) SEM images of the GNS array at varying O₂ plasma etching times; scale bars are 200 nm for the main images and 100 nm for the insets. (c) Height profiles of the GNS array at different O₂ plasma etching times. (d) Variation in the width and length of GNSs as a function of O₂ plasma etching time. (e) Large-scale SEM image of a nearly perfect GNS array; scale bar is 2 μm. The probabilities of finding interstitials and vacancies per unit cell are defined as P_{inter} and P_{vacancy} , respectively. The relative standard deviation σ_{relative} for size distribution is the absolute particle size standard deviation relative to the large area. (f) EDS maps of different intermetallic alloy arrays: PdAu, PdPt, PtFe, and FePd NSs.

a reduction in the y -direction of the stamps.³⁷ Consequently, the application of a uniaxial strain on the PDMS stamp transformed the pattern holes from circular to elliptical shape. The stretchable PDMS stamp was subsequently inked with a 1 wt % solution of poly(4-vinylpyridine) (P4VP) in isopropanol and brought into conformal contact with a substrate to generate periodic nanocylindroid arrays (step 2) (Figures S1 and S2). A brief O₂ plasma descum process was used to remove excess P4VP between nanocylindroids to ensure that negatively charged metal precursors (e.g., AuCl₄[−], PdCl₄^{2−}, PtCl₄^{2−}, and so forth) were accurately absorbed into the

nanocylinders through electrostatic interactions (Figure S3). Subsequently, metal ions loaded in the polymeric nanodomains were reduced to the zero valence state through free electrons generated by the O₂ plasma treatment (step 3). Finally, the resulting nanoclusters were thermally annealed at 500 °C to yield a highly uniform inorganic NS array with high positional precision and uniform orientation (step 4) (Figure S3).

Representative scanning electron microscopy (SEM) image of the GNS array in Figure 1b indicates the high uniformity of GNSs and high-precision directional orientation. The GNS array shows a rectangular lattice periodicity ($P_x = 550$ nm, $P_y =$

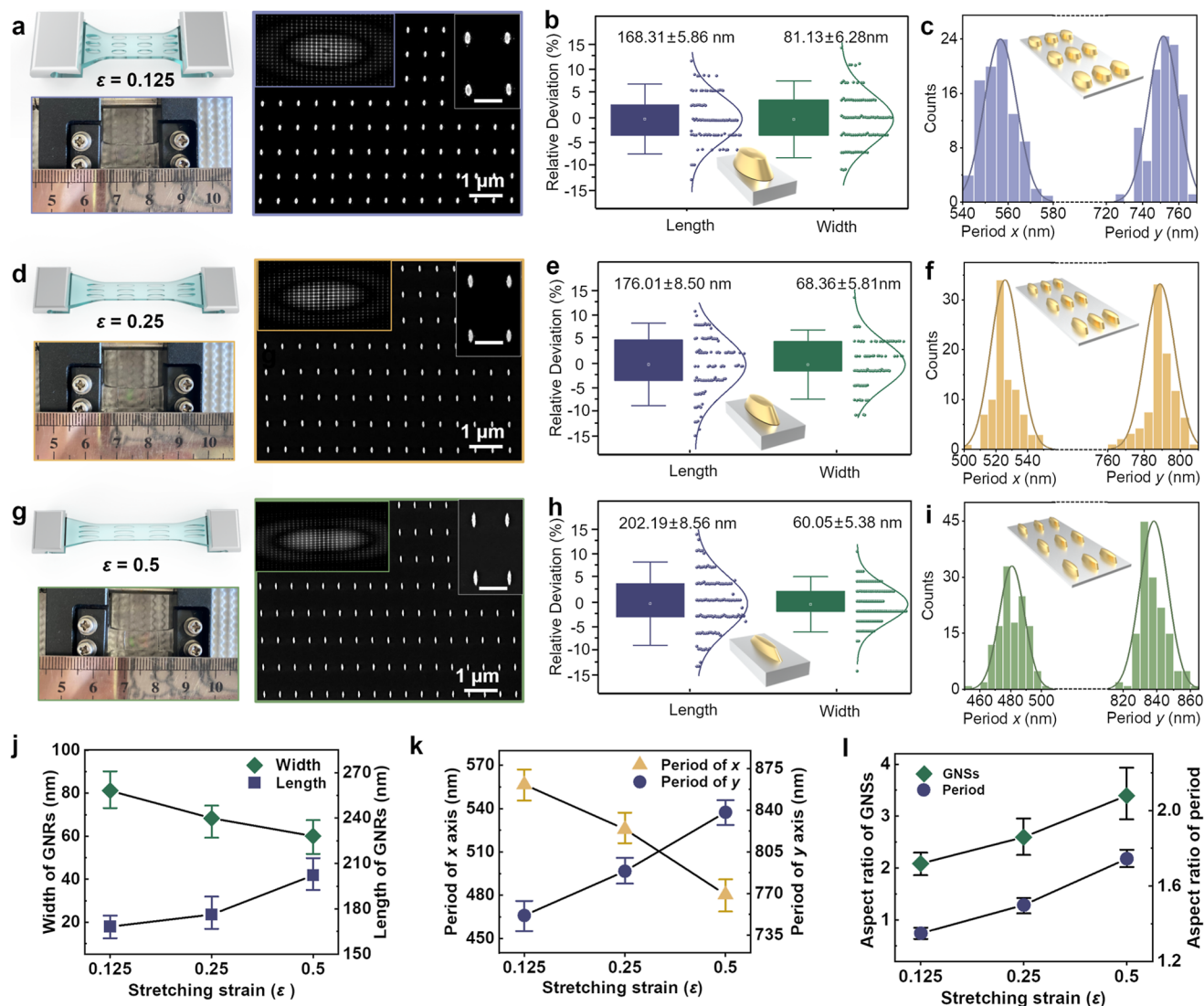


Figure 3. Tunable lattice periodicity and size of GNS array by a stretching elastomeric substrate in soft lithography. (a,d,g) Photograph of the PDMS patterns and SEM images of GNS array in different mechanical strains (ϵ). Insets at the left are FFT patterns of GNS array. The right insets are high magnification SEM images of the GNS array, the scale bar is 400 nm. The statistical distributions of the GNS size (b,e,h) and lattice periodicity (c,f,i) of the array corresponding to figure (a,d,g), respectively. The variation in the width, length (j), lattice periodicity (k), and aspect (l) of GNS array as a function of mechanical strains.

800 nm) due to the mechanical deformation of the originally square-like lattice of the master. The GNS' cross-section along its longitudinal and transverse axis displays a dome-like and cylindrical shape, respectively (inset in Figures 1b and S4). The anisotropic geometry of the GNS was further confirmed by our atomic force microscopy (AFM) measurements (Figure 1c). The height (h) of GNSs was ~ 160 nm and the length and width were ~ 210 and ~ 100 nm. Dark field microscopy image of the GNS array is presented in Figure 1d, where each individual blue dot represents a single GNS. The uniform blue scattering across the array indicates effective large-scale control of the orientational order. This centimeter-scale array displayed a clear structure color, indicating the high uniformity of the overall structures (Figure 1d inset). The GNS array exhibited a narrower SLR line width than that of the LSPR and a much higher Q -factor. Incidentally, the inset field profiles in Figure 1e also reveal that the SLR provides a more significant field enhancement, with $|E_{\max}(\lambda_{\text{SLR}})|^2 \sim 33 |E_{\max}(\lambda_{\text{LSPR}})|^2$. The measured transmission spectrum in Figure 1f shows an SLR

line width of ~ 3.9 nm and a Q -factor of ~ 309 , which is in good agreement with the predicted spectrum. It is worth noting that dimethyl sulfoxide (DMSO) was used to replace air as the superstrate, carefully matched to the substrate material to ensure a symmetric refractive index environment, as asymmetric refractive index environments have been shown to suppress SLR due to the additional reflection produced by the substrate.^{13,38}

During the fabrication process of GNS array, thermal annealing temperature and O_2 plasma etching time play crucial roles in its morphology and size. The morphology evolution of NPs at different thermal annealing temperatures is exhibited in Figure 2a. At 500°C , the spindle-shaped nanoclusters, initially composed of loose small NPs, transform into a spindle structure with tightly packed particles. Although the morphology remains the same, the size is reduced compared to that of the initial state. When the sintering temperature was further increased to 700°C , the length of NSs was decreased, eventually evolved into nanospheres in 1000°C (Figure S5).

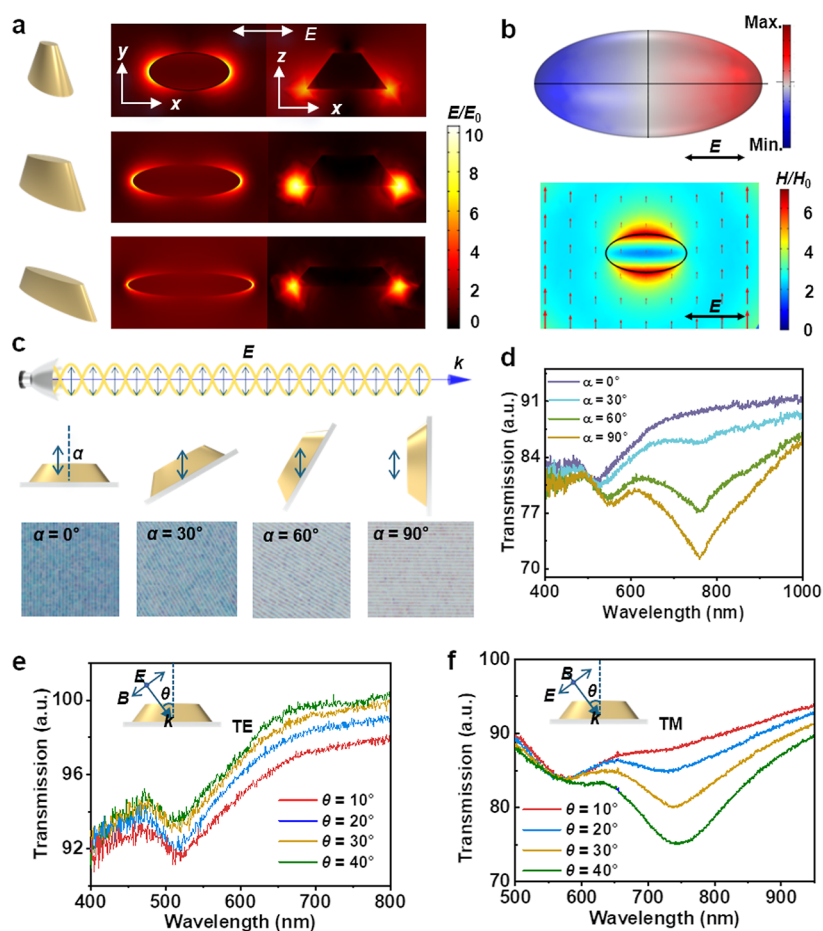


Figure 4. Polarization-dependent optical properties of GNS. Simulated electric field distribution (a) and charge distribution (b) of the GNS at the resonance wavelength with different strains. Bright-field photographs (c) and experimental transmission spectra (d) of the GNS array under varied polarized angles. The light was a tungsten-halogen lamp through a linear polarizer. Transmission spectra of incident light at different angles under TE (e) and TM (f) polarization conditions.

The shape evolution is primarily driven by the surface energy. In the initial stages of high-temperature sintering, the Ostwald ripening effect predominates. Ostwald ripening describes the process where smaller particles dissolve while larger particles grow, leading to a reduction in the total free energy.^{39–42} This process can be described by the following equation

$$\frac{dR}{dt} = k \left(\frac{1}{R} - \frac{1}{R_{\text{crit}}} \right) \quad (1)$$

where R is the radius of the particle; t is the time; k is the rate constant for diffusion; and R_{crit} is the critical radius. As time progresses, smaller particles dissolve, allowing larger particles to continue growing. Further increase of the sintering temperature, the driving force for the shape change is the reduction of total free energy. The total free energy of the system can be expressed as

$$G_{\text{total}} = G_v + G_s \quad (2)$$

because the volume free energy G_v is independent of shape, while the surface free energy G_s is proportional to surface area, the system minimizes its total free energy by reducing the surface area, leading to a spherical shape.

Beyond varying the annealing temperature, different oxygen plasma treatment times (t_p) determine different morphologies

and sizes of GNS arrays (Figure 2b). At $t_p = \sim 10$ s, the excess P4VP outside the nanocylindroid arrays was not effectively removed. The residual of P4VP outside the patterns caused the formation of many tiny NPs surrounding the NSs, like a satellite structure after thermal annealing. At $t_p = \sim 15$ s, the number of tiny NPs surrounded by NSs decreased due to the decrease in the P4VP residue. As the plasma treatment time was extended to 20 s, the tiny NPs outside the nonspecific patterns were efficiently suppressed owing to the effective removal of undesired P4VP (Figure S6). Moreover, we observed that the size of the GNSs can be accurately controlled via oxygen plasma treatment. Increasing the oxygen plasma treatment time led to an approximately linear reduction in the height and diameter of the GNSs, while maintaining the anisotropy and uniformity of the nanostructures. When the duration of plasma treatment increased from 10 to 15, 20, and 25 s, the height of the GNSs decreased from 206 ± 1.2 to 181 ± 1.5 , 160 ± 1.9 , and eventually to 130 ± 2.4 nm, respectively (Figure 2c and Figure S7). Accordingly, this observation is accompanied by a decrease in the transverse and longitudinal dimensions of the GNSs, caused by the decrease in the size of P4VP nanocylindroids (Figures 2d and S8). The small standard deviations of size and nearly zero defects of nano-objects over a large area confirm the high uniformity of GNS array (Figure 2e). This method can be extended to fabricate intermetallic alloy arrays from different metal precursors. In

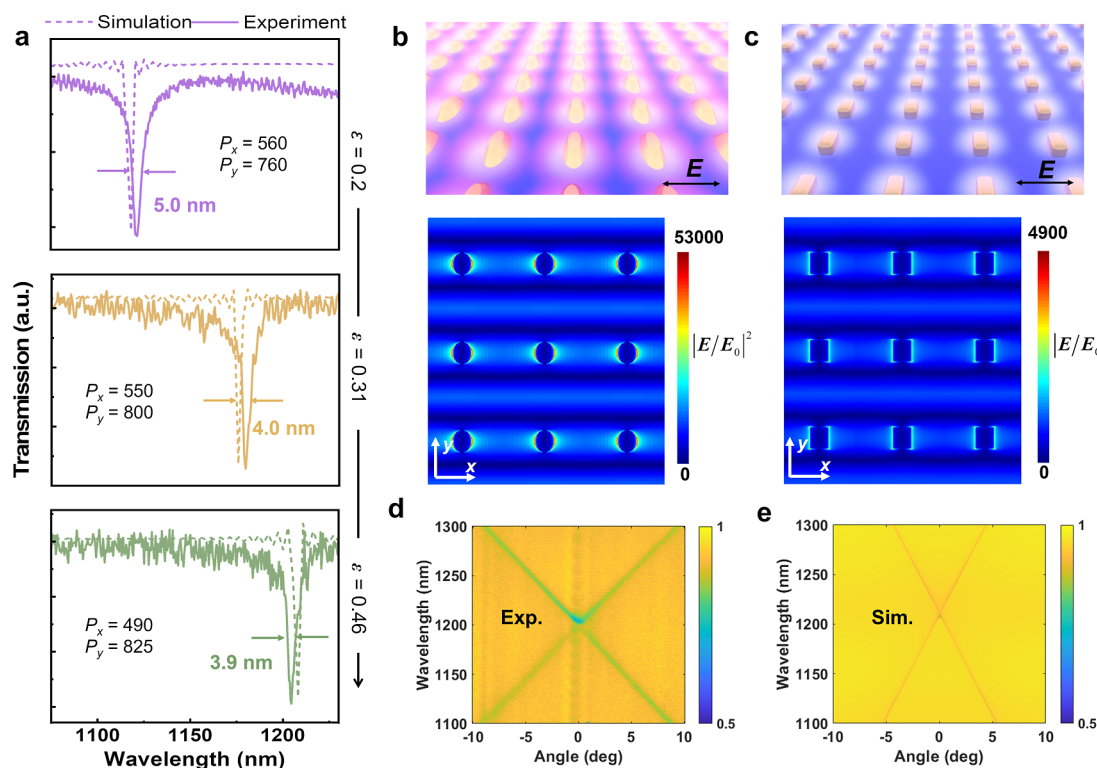


Figure 5. Mechano-tunable optical properties of GNS rectangle arrays. (a) Measured and calculated (dash line) transmission spectra of the GNS array with different periodicities under x -polarization: $P_x = 560$ nm and $P_y = 760$ nm (purple line), $P_x = 550$ nm and $P_y = 800$ nm (yellow line), and $P_x = 490$ nm and $P_y = 825$ nm (green line). FDTD-calculated near-field distribution map of $|E/E_0|^2$ of GNS array (b) and anisotropic structures prepared by e-beam lithography (c) at band-edge lattice plasmons, respectively. ($\lambda = 1120$ nm, $\theta = 0^\circ$). (d) Experimental and (e) simulated dispersion diagrams of GNP arrays under TE polarization.

Figure 2f, energy-dispersive spectroscopy (EDS) mapping of a typical intermetallic alloy nanostructure array comprising PdAu, PdPt, PtFe, and FePd is presented. The SEM images indicate that all samples exhibit high-quality array characteristics (Figure S9).

Mechanical stretching was used to effectively modulate the size and the periodicity of GNSs. Applying different strains (ϵ) in the uniaxial direction yielded GNSs with controllable aspect ratios and varied lattice constant. Three different ϵ values of 0.125, 0.25, and 0.5 were chosen for the demonstration of concept. Figure 3a,d,g shows the photographs of stretched devices and the corresponding SEM images of well-ordered arrays over a large area. The fast Fourier transform (FFT) patterns of the resulting arrays exhibit a sharp rectangular shape, indicating a high precision directional orientation of the GNSs in the arrays. The magnified SEM image shows changes in the GNS size and period. To better present the varied results of the well-defined structure, the statistical distributions of the size and center distance of the GNSs are presented in Figure 3. As the applied strain increased, the longitudinal size of GNSs had a linear increase from 168 ± 7 to 176 ± 9 and to 202 ± 9 nm, respectively. Meanwhile, the transverse axis decreased from 81 ± 7 to 68 ± 6 nm and ultimately to 60 ± 4 nm, due to a Poisson's ratio of 0.5 (Figure 3b,e,h,j,l). Accordingly, this observation is accompanied by an increase in period of y (P_y), which is 750 ± 11 , 800 ± 8 , and 840 ± 12 nm, respectively, and a decrease in period of x (P_x) of 555 ± 10 , 525 ± 8 , and 480 ± 11 nm, respectively (Figure 3c,f,i,k,l). The relative standard deviation of size demonstrates the high consistency of the structural units and their good alignment

within the array. The adjustability of the aspect ratio of GNS size and period is favorable to modulate the optical properties.

Optical Properties of GNS Array. To demonstrate the high-precision directional orientation of the GNS array, we further characterized the polarization-dependent optical properties of centimeter-sized arrays. Figure 4a,b shows the calculated electromagnetic field distribution and charge distribution. Due to the unique spindle-shaped structure, the electric field distribution is primarily concentrated at the bottom tip, and as the stretching increases, the electric field at the tip becomes stronger. Figure 4c shows the camera-captured bright-field images of the GNS array under a varied polarized angle (α). From 0 to 90° , the color of the array is uniformly modulated from green to red. The uniform color change over the entire array indicates that the GNSs were all oriented in one direction. The varied colors can be explained through transmission spectra of the GNS array (Figure 4d). At 0° , the polarizer is orthogonal to the longitudinal axis of the GNS and, therefore, the spectra exhibit only one transverse transmission peak around 525 nm, causing the GNSs to appear green. As the polarized angle increases, the intensity of the long-axis transmission peak gradually increases. Upon $\alpha = 90^\circ$, the transverse peak still exists and undergoes a redshift, considering that the GNSs consist of small particles. But the perceived color of GNS array is dominated by the longitudinal surface plasmon oscillations and appear red (760 nm). Figure 4e,f presents the transmission spectra from GNSs for a series of incident angles under TE and TM polarization. Under TE-polarized light illumination, as the incident angle increases from 10 to 40° , the electric field direction consistently remains parallel to the short axis. Consequently, a transverse LSPR

Table 1. Summary of Experimentally Obtained Q-Factors of SLR in AuNP Arrays

Q-factor	fwhm (nm)	E^2/E_0^2	λ (nm)	material	light source	area (μm^2)	reference
309	3.9	53,000	1205	Au NSs	tungs.-halogen lamp	7000 $\times$ 5000	This work
25	26.6	2000	930	Au disk	collimated source	135 $\times$ 135	49
150	5	1000	750	Au NPs	tungs.-halogen lamp	18 cm^2	50
215	4	5000	867	Au NPs	tungs.-halogen lamp	1000*1000	15
218	4	2500	870	Au NPs	tungs.-halogen lamp	large	41
2340	0.66	2025	1550	Au rectangular NPs	supercontinuum	600*600	21
134	6.5	800	870	Au NPs	tungs.-halogen lamp	18*18	51
110	6	500	886	Au nanocrescent	tungs.-halogen lamp	1000*1000	52
102	6	*	613	Au NPs	tungs.-halogen lamp	NA	53
300	5	*	1500	Au/graphene	tungs.-halogen lamp	300*100	33
270	4.1	*	1150	Au NPs	supercontinuum	7000*7000	54

peak appears solely at 525 nm in the transmission spectra, and it remains unchanged with varying incident angles. Under TM-polarized light illumination, in addition to the component parallel to the long axis of GNSs, the electric field also exhibits a component parallel to the Z-axis. Moreover, with an increase in the incident angle, the intensity of the electric field parallel to the Z-axis becomes stronger. Due to the close proximity of the GNS dimensions in the Z-direction and along the long axis, the wavelength of its optical response closely aligns with the longitudinal mode of LSPR at 760 nm. Therefore, as the incident angle increases from 10 to 40°, the enhancement of the longitudinal LSPR mode is observed, showcasing the polarization-dependent characteristic of the GNS array.

Benefiting from the tunable lattice periodicity and size of the GNS array, we achieved a modulated SLR wavelength in the near-infrared region with varying strains in soft lithography (Figure 5). Unlike a square lattice, a rectangular lattice with two different lattice constants in the x - and y -directions, resulting in two different SLR modes. We defined SLR- x as the mode induced by polarization in the x -direction, SLR- y induced by polarization in the y -direction. The measured and calculated SLR- x transmission spectra of GNS rectangle arrays with different lattice periods and sizes were characterized by immersing the arrays in DMSO ($n = 1.48$), with refractive index matching that of the quartz (Figure 5a). We compared the optical responses of the GNS array under three different strains. The array exhibited a SLR- x mode with a full width at half-maximum of 5 nm at $\lambda_{\text{SLR-}x} = 1120$ nm, as the applied strain $\varepsilon = 0.2$, with lattice constant $P_y = 760$ nm. As the increase of lattice constant P_y ($P_y = 800/825$ nm) due to the increase in strain, the peak positions of SLR- x undergo a significant red shift, with $\lambda_{\text{SLR-}x}$ being 1180 and 1205 nm, respectively. Additionally, the line widths are slightly narrower. The quality factor also increased significantly from $Q = 224$ –309. A relatively high- Q resonance at $\lambda_{\text{SLR-}x}$ is contributed by the geometry of individual nanostructure and the period of the lattice cooperatively.^{21,43} The optical response of a nanostructure can be approximated using the polarizability of a Lorentzian dipole

$$\alpha(\omega) = \frac{A_0}{\omega - \omega_0 + i\gamma} \quad (3)$$

where A_0 is the oscillator strength, $\omega_0 = 2\pi c/\lambda_{\text{LSPR}}$ corresponds to the NP resonance frequency and γ is the damping term. The contribution of the particle lattice to the polarizability can be introduced using the LSA⁴⁴

$$\alpha^*(\omega) = \frac{\alpha(\omega)}{1 - \varepsilon_0 \alpha(\omega) S(\omega)} \quad (4)$$

where $\alpha^*(\omega)$ is known as the effective polarizability of the entire arrays and $S(\omega)$ corresponds to the lattice sum. This latter term depends only on the arrangement of the lattice. Recent theoretical studies show the nanostructure geometry mainly dictates its coupling efficiency to free space, while the spectral location of the SLR wavelength is mainly dictated by the lattice period and the background index $\lambda_{\text{SLR}} \approx n \times P$.^{21,45} As a consequence, although tunable strain leads to changes in the nanostructural dimensions and lattice constants, the variable lattice parameters play a key role in regulating the SLR peak position. The experimental results are consistent with our FDTD simulation of transmission spectra of the arrays of NS with different lattice constants (Figure 5a). The SLR- x line width is narrower, corresponding to a much higher Q-factor. Incidentally, the SLR- x provides a more significant field enhancement, which is 33 times that of LSPR around GNSs in the array, and is much larger than that around single NP at λ_{LSPR} (Figures 5c and S10). In Figure 5b,c, we compared the electric field of GNSs with that of the structure prepared by EBL and found that the electric field at the lattice resonance response of our structure is much higher than that of the structure prepared by e-beam lithography (Table 1). Experimental dispersion diagrams of GNS array under TE polarization showed a clear band-edge lattice mode in the near-infrared region that could support potential applications in nanolasing and sensing, which is in good agreement with our simulations (Figures 5d,e and S11).

Different from SLR- x , the SLR- y induced by polarization in the y -direction is at lower wavelengths (Figure S12). This phenomenon is explained by excitation along the y -axis inducing electron oscillations in the parallel orientation for each NS. However, these dipoles radiate the scattered light mainly perpendicular to the dipole moment (Figure S13). As a consequence, the SLR mode is dictated by the lattice constant, which is normal to the incident polarization for an asymmetric lattice.⁴⁶ Note that the $\lambda_{\text{SLR-}y}$ is blue-shifted unlike the $\lambda_{\text{SLR-}x}$ shifts toward higher wavelengths because of the strain-induced nondegeneration. Simultaneously, line width of $\lambda_{\text{SLR-}y}$ is broader as compared to $\lambda_{\text{SLR-}x}$ because of the reduced plasmonic damping of gold for larger wavelengths in the visible wavelength range.⁴⁷ Additionally, a low-quality resonance can be obtained due to the significant overlap between the broad LSPR mode and SLR.⁴⁸

CONCLUSIONS

In conclusion, we have developed a robust approach for fabricating high-quality anisotropic inorganic NS arrays of precisely oriented NS with tunable optical responses. This fabrication approach shows at least the following advantages: (i) it is simple to implement for scalable creation of NS arrays not requiring special tools; (ii) it is applicable to tune the sizes, lattice constants, compositions, and potentially different shapes of the NPs; and (iii) it enables one to access tunable optical resonance and optical anisotropy. The GNS array prepared by our approach exhibited a ultranarrow SLR with the Q -factor of 309 at the wavelength 1205 nm and strong electric field enhancement ($|E/E_0|^2 = 53,000$), under the symmetric refractive index environment. Moreover, this method can be potentially extended to fabricate plasmonic arrays with more complex nanostructures (e.g., polyhedral inorganic arrays). And we expect to further improve the optical cavity, enabling the excitation of SLR in asymmetric dielectric environments, which could lead to practical applications such as optofluidic sensors⁵⁵ and molecule detection.³⁴

METHODS

Materials. Metal compounds [(HAuCl₄·3H₂O), K₃Fe (CN)₆, Na₂PdCl₄, K₃Co (CN)₆, and K₂PtCl₄], fluorosilane (tridecafluoro-1,1,2,2-tetrahydrooctyl-1-trichlorosilane), and polymer P4VP (average $M_w \sim 160,000$) were purchased from Sigma-Aldrich. Sulfuric acid (H₂SO₄), hydrogen peroxide (H₂O₂), hydrochloric acid (HCl), ethanol, acetone, and 2-propanol were purchased from Sinopharm Chemical Reagent. The relief nanopatterned silicon masters (square array of posts with diameters of $d = 250$ nm and $d = 190$ nm, height $h = 350$ nm, and pitch $a_0 = 600$ nm) were purchased from Max Levy Autograph, Inc. (MLA). All of the chemicals and reagents were used as received. Deionized water used in the experiments was purified using a Millipore water purification system with a minimum resistivity of 18.2 MΩ·cm.

Preparation of the Substrate. Silicon (Si) wafers or quartz slides were sonicated in ethanol, acetone, and deionized water for 5 min and then immersed in a H₂SO₄ (98%)/H₂O₂ (30%) = 7:3 (v/v) mixture of piranha solution. The surface was hydroxylated by boiling for 30 min, and the residual acid was removed by rinsing the Si wafers with deionized water.

Patterning P4VP Nanocylindroid on the Substrate. The relief nanopatterned silicon masters were passivated with a fluorosilane (tridecafluoro-1,1,2,2-tetrahydrooctyl-1-trichlorosilane) for 24 h in a vacuum desiccator. SYLGARD 184-PDMS (base and curing agents are mixed in a 10:1 mass ratio) was poured on the masters and cured for 2 h at 80 °C. The obtained PDMS molds were placed in the stretching device and a certain stress was applied to obtain the desired pattern. 50 μL of P4VP (1 wt %) isopropanol solutions was dropped on the surface of the mold, and then the processed substrate was immediately placed into conformal contact with the mold. After the solvent evaporated for 1 h, the mold was removed, and P4VP posts were formed on the substrate.

Fabrication of the GNS Arrays. The P4VP nanocylindroid arrays were treated by oxygen plasma at 60 mTorr, 35 s.c.c.m., and 60 W of power for 5, 10, 15, 20, and 25 s. For GNSs, the P4VP arrays were immersed in 3 mM HAuCl₄ aqueous solution for 1 h to deposit gold ionic precursors into P4VP posts. After precursor loading, the samples were rinsed with deionized water and dried with a N₂ gas. Then, oxygen plasma treatment for 2 min completely removed the P4VP and ordered gold nanoclusters were left on the substrate. Finally, the gold nanocluster array was annealed in 500 °C with air in 2 h to obtain monodisperse GNS arrays.

Fabrication of Other Inorganic NS Arrays. For the PdAu, PdPt, PtFe, and FePd alloy arrays, the P4VP arrays were separately immersed in 1 mM Na₂PdCl₄, 1 mM HAuCl₄, 1 mM Na₂PdCl₄, 1 mM K₂PtCl₄, 1 mM K₂PtCl₄ and 1 mM K₃Fe (CN)₆ solutions, each

in a 1:1 volume ratio, for 30 min. All precursor salt solutions were prepared by using a 3% HCl solution. Then, plasma treatment for 2 min reduced the metal precursors, forming alloy clusters. Subsequently, the sample was annealed at 500 °C for 1 h under an Ar atmosphere to obtain the alloy arrays.

Characterization. Samples on the substrate were directly imaged by using a Gemini Ultra55 (Zeiss) SEM. The height of P4VP and GNP arrays and the film thickness were characterized using Multimode 8 (Bruker) AFM. The polarization-angle-dependent transmission spectra of GNS arrays were measured by an angular resolution spectral system R1 (Ideaoptics). The transmission and dispersion characteristics of GNS arrays were investigated by using a custom-built system for measuring momentum-space with polarization resolution. In this setup, the sample was subjected to Kohler illumination, where it was simultaneously illuminated with light at various incident angles. Subsequently, the light transmitted through the sample was gathered by an objective lens with a numerical aperture of 0.4 and directed to a spectrometer positioned at the Fourier plane. Prior to being entered into the spectrometer, the transmitted light passed through a linear polarizer. This configuration allowed for the collection of polarization-resolved dispersion data, which were then plotted in wavelength (λ) versus incident angle (θ) units.

Numerical Simulation. Numerical simulations of far-field extinction spectra and near-field electric field profiles were performed using a commercial software package, Ansys Lumerical FDTD three-dimensional (3D) electromagnetic solver. Periodic boundary conditions were set along both X and Y directions, and perfectly matched layers (20 layers) were employed along the Z direction. The light source was set as a plane wave with a wavelength range of 500–1300 nm and was injected along the z axis, which was normal incidence. A frequency domain power monitor was placed on the opposite side of the substrate to record the transmittance data. The gold permittivity was modeled according to the experimental data reported by Johnson and Christy. The structure of the nanocylindroid was enclosed by a mesh box with a mesh step of 2 nm. The background material was set as SiO₂ due to extremely sharp spectra features resulting from SLRs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.4c18423>.

Characterization of P4VP NS arrays; characterization of GNS arrays; characterization of other inorganic NS arrays; optical properties of GNS arrays; and schematic graph of optical measurement system setup (PDF)

AUTHOR INFORMATION

Corresponding Authors

Fan Yang — School of Physical Science and Technology, Tiangong University, 300387 Tianjin, China; orcid.org/0000-0002-9395-5310; Email: yangfan@tiangong.edu.cn

Zhihong Nie — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China; orcid.org/0000-0001-9639-905X; Email: znief@fudan.edu.cn

Authors

Hongyan Li — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China

Jingyi Zhao — Department of Physics, Key Laboratory of Micro- and Nano-Photonic Structures (Ministry of Education), and State Key Laboratory of Surface Physics, Fudan University, 200433 Shanghai, China

Yazi Wang — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China

Haitao Liu — School of Physical Science and Technology, Tiangong University, 300387 Tianjin, China

Qianyun Chen — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China

Yilin Bao — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China

Miaoen Zhou — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China

Yue Li — School of Physical Science and Technology, Tiangong University, 300387 Tianjin, China; orcid.org/0000-0002-1202-4224

Yutao Sang — State Key Laboratory of Molecular Engineering of Polymers, Department of Macromolecular Science, Fudan University, 200433 Shanghai, China; orcid.org/0000-0003-1170-7362

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsnano.4c18423>

Author Contributions

H.L. performed the theoretical analysis. H.L. fabricated the samples. H.L. and F.Y. carried out the numerical simulation. H.L. and J.Z. carried out the optical measurements. Y.W., Y.L., and Y.S. took part in discussions and analysis. H.L. wrote the manuscripts. H.L. and Q.C. participated in the revision of the article figure. F.Y. and Z.N. supervised the project. All authors contributed to the manuscript and approved the final version.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge the financial support from the Innovation Program of Shanghai Municipal Education Commission (2021-01-07-00-E00073), Shanghai Municipal Science and Technology Project (23JC1400100), and the National Natural Science Foundation of China (grants 5210030633, 52125308, 91963107, 51973038, 52103077, and 52203327). The authors thank Shunsheng Ye, Wenhao Dong for their support on fabrication and characterization, and thank Nanofabrication Lab at Fudan university and Ideaoptics Technology Ltd., China for facility support.

REFERENCES

- (1) Hu, M.; Chen, J.; Li, Z.-Y.; Au, L.; Hartland, G. V.; Li, X.; Marquez, M.; Xia, Y. J. C. S. R. Gold nanostructures: engineering their plasmonic properties for biomedical applications. *Chem. Soc. Rev.* **2006**, *35* (11), 1084–1094.
- (2) Rycenga, M.; Cobley, C. M.; Zeng, J.; Li, W.; Moran, C. H.; Zhang, Q.; Qin, D.; Xia, Y. Controlling the Synthesis and Assembly of Silver Nanostructures for Plasmonic Applications. *Chem. Rev.* **2011**, *111* (6), 3669–3712.
- (3) Anker, J. N.; Hall, W. P.; Lyandres, O.; Shah, N. C.; Zhao, J.; Van Duyne, R. P. J. N. m. Biosensing with plasmonic nanosensors. *Nat. Mater.* **2008**, *7* (6), 442–453.
- (4) Qiu, G.; Gai, Z.; Tao, Y.; Schmitt, J.; Kullak-Ublick, G. A.; Wang, J. J. A. n. Dual-functional plasmonic photothermal biosensors for highly accurate severe acute respiratory syndrome coronavirus 2 detection. *ACS Nano* **2020**, *14* (5), 5268–5277.

- (5) Altug, H.; Oh, S.-H.; Maier, S. A.; Homola, J. J. N. n. Advances and applications of nanophotonic biosensors. *Nat. Nanotechnol.* **2022**, *17* (1), 5–16.
- (6) Zhou, L.; Swearer, D. F.; Zhang, C.; Robatjazi, H.; Zhao, H.; Henderson, L.; Dong, L.; Christopher, P.; Carter, E. A.; Nordlander, P.; Halas, N. J. Quantifying hot carrier and thermal contributions in plasmonic photocatalysis. *Science* **2018**, *362* (6410), 69–72.
- (7) Neubrech, F.; Duan, X.; Liu, N. J. S. A. Dynamic plasmonic color generation enabled by functional materials. *Sci. Adv.* **2020**, *6* (36), No. eabc2709.
- (8) Song, M.; Feng, L.; Huo, P.; Liu, M.; Huang, C.; Yan, F.; Lu, Y.-q.; Xu, T. J. N. N. Versatile full-colour nanopainting enabled by a pixelated plasmonic metasurface. *Nat. Nanotechnol.* **2023**, *18* (1), 71–78.
- (9) Autere, A.; Jussila, H.; Dai, Y.; Wang, Y.; Lipsanen, H.; Sun, Z. J. A. M. Nonlinear optics with 2D layered materials. *Adv. Mater.* **2018**, *30* (24), 1705963.
- (10) Kauranen, M.; Zayats, A. V. J. N. p. Nonlinear plasmonics. *Nat. Photonics* **2012**, *6* (11), 737–748.
- (11) Tittel, A.; Leitis, A.; Liu, M.; Yesilkoy, F.; Choi, D.-Y.; Neshev, D. N.; Kivshar, Y. S.; Altug, H. J. S. Imaging-based molecular barcoding with pixelated dielectric metasurfaces. *Science* **2018**, *360* (6393), 1105–1109.
- (12) Koshelev, K.; Kruk, S.; Melik-Gaykazyan, E.; Choi, J.-H.; Bogdanov, A.; Park, H.-G.; Kivshar, Y. J. S. Subwavelength dielectric resonators for nonlinear nanophotonics. *Science* **2020**, *367* (6475), 288–292.
- (13) Kravets, V. G.; Kabashin, A. V.; Barnes, W. L.; Grigorenko, A. N. J. C. r. Plasmonic surface lattice resonances: a review of properties and applications. *Chem. Rev.* **2018**, *118* (12), 5912–5951.
- (14) Zhang, X.; Liu, Y.; Han, J.; Kivshar, Y.; Song, Q. J. S. Chiral emission from resonant metasurfaces. *Science* **2022**, *377* (6611), 1215–1218.
- (15) Deng, S.; Li, R.; Park, J.-E.; Guan, J.; Choo, P.; Hu, J.; Smeets, P. J.; Odom, T. W. J. Ultranarrow plasmon resonances from annealed nanoparticle lattices. *Proc. Natl. Acad. Sci. U.S.A.* **2020**, *117* (38), 23380–23384.
- (16) Kim, I.; Kim, W.-S.; Kim, K.; Ansari, M. A.; Mehmood, M. Q.; Badloe, T.; Kim, Y.; Gwak, J.; Lee, H.; Kim, Y.-K. J. S. A.; et al. Holographic metasurface gas sensors for instantaneous visual alarms. *Sci. Adv.* **2021**, *7* (15), No. eabe9943.
- (17) Zou, S.; Janel, N.; Schatz, G. C. J. T. J. o. c. p. Silver nanoparticle array structures that produce remarkably narrow plasmon lineshapes. *J. Chem. Phys.* **2004**, *120* (23), 10871–10875.
- (18) Khlopin, D.; Laux, F.; Wardley, W. P.; Martin, J.; Wurtz, G. A.; Plain, J.; Bonod, N.; Zayats, A. V.; Dickson, W.; Gérard, D. J. J. B. Lattice modes and plasmonic linewidth engineering in gold and aluminum nanoparticle arrays. *J. Opt. Soc. Am. B* **2017**, *34* (3), 691–700.
- (19) Rodriguez, S.; Schaafsma, M.; Berrier, A.; Gómez Rivas, J. Collective resonances in plasmonic crystals: Size matters. *Phys. B* **2012**, *407* (20), 4081–4085.
- (20) Le-Van, Q.; Zoethout, E.; Geluk, E. J.; Ramezani, M.; Berghuis, M.; Rivas, J. G. Enhanced quality factors of surface lattice resonances in plasmonic arrays of nanoparticles. *Adv. Opt. Mater.* **2019**, *7* (6), 1801451.
- (21) Bin-Alam, M. S.; Reshef, O.; Mamchur, Y.; Alam, M. Z.; Carlow, G.; Upham, J.; Sullivan, B. T.; Ménard, J. M.; Huttunen, M. J.; Boyd, R. W. J. N. c.; et al. Ultra-high-Q resonances in plasmonic metasurfaces. *Nat. Commun.* **2021**, *12* (1), 974.
- (22) van der Hoeven, J. E.; Gurunaryanan, H.; Bransen, M.; de Winter, D. M.; de Jongh, P. E.; van Blaaderen, A. J. A. F. M. Silica-coated gold nanorod supraparticles: a tunable platform for surface enhanced Raman spectroscopy. *Adv. Funct. Mater.* **2022**, *32* (27), 2200148.
- (23) Lohse, S. E.; Murphy, C. J. J. C. o. M. The quest for shape control: a history of gold nanorod synthesis. *Chem. Mater.* **2013**, *25* (8), 1250–1261.

- (24) Wang, M.; Gao, C.; He, L.; Lu, Q.; Zhang, J.; Tang, C.; Zorba, S.; Yin, Y. J. o. t. A. C. S. Magnetic tuning of plasmonic excitation of gold nanorods. *J. Am. Chem. Soc.* **2013**, *135* (41), 15302–15305.
- (25) Hakala, T. K.; Moilanen, A. J.; Väkeväinen, A. I.; Guo, R.; Martikainen, J.-P.; Daskalakis, K. S.; Rekola, H. T.; Julku, A.; Törmä, P. J. N. Bose–Einstein condensation in a plasmonic lattice. *Nat. Phys.* **2018**, *14* (7), 739–744.
- (26) Wang, H. C.; Martin, O. J. F. Polarization-Controlled Chromo-Encryption. *Adv. Opt. Mater.* **2023**, *11* (7), 2202165.
- (27) Taylor, A. B.; Michaux, P.; Mohsin, A. S.; Chon, J. W. J. O. e. Electron-beam lithography of plasmonic nanorod arrays for multi-layered optical storage. *Opt. Express* **2014**, *22* (11), 13234–13243.
- (28) Zhang, H.; Cadusch, J.; Kinnear, C.; James, T.; Roberts, A.; Mulvaney, P. J. A. n. Direct assembly of large area nanoparticle arrays. *ACS Nano* **2018**, *12* (8), 7529–7537.
- (29) Wang, Y.; Li, H.; Chu, J.; Xia, Y.; Ye, S.; Yang, F.; Cao, W.; Ge, J.-Y.; Xu, Y.; Zhu, M.; Pan, H.; Nie, Z. Site-Selective Assembly of Centimeter-Scale Arrays of Precisely Oriented Magnetic Nanoparticles. *ACS Nano* **2022**, *16* (12), 21208–21215.
- (30) Zhang, H.; Liu, Y.; Shahidan, M. F. S.; Kinnear, C.; Maasoumi, F.; Cadusch, J.; Akinoglu, E. M.; James, T. D.; Widmer-Cooper, A.; Roberts, A. J. A. F. M.; et al. Direct assembly of vertically oriented, gold nanorod arrays. *Adv. Funct. Mater.* **2021**, *31* (6), 2006753.
- (31) Flauraud, V.; Mastrangeli, M.; Bernasconi, G. D.; Butet, J.; Alexander, D. T.; Shahabi, E.; Martin, O. J.; Brugger, J. J. N. n. Nanoscale topographical control of capillary assembly of nanoparticles. *Nat. Nanotechnol.* **2017**, *12* (1), 73–80.
- (32) Yang, A.; Hoang, T. B.; Dridi, M.; Deeb, C.; Mikkelsen, M. H.; Schatz, G. C.; Odom, T. W. Real-time tunable lasing from plasmonic nanocavity arrays. *Nat. Commun.* **2015**, *6* (1), 6939.
- (33) Thackray, B. D.; Thomas, P. A.; Auton, G. H.; Rodriguez, F. J.; Marshall, O. P.; Kravets, V. G.; Grigorenko, A. N. Super-Narrow, Extremely High Quality Collective Plasmon Resonances at Telecom Wavelengths and Their Application in a Hybrid Graphene-Plasmonic Modulator. *Nano Lett.* **2015**, *15* (5), 3519–3523.
- (34) Chen, Z.; Cao, A.; Liu, D.; Zhu, Z.; Yang, F.; Fan, Y.; Liu, R.; Huang, Z.; Li, Y. Self-Confined Dewetting Mechanism in Wafer-Scale Patterning of Gold Nanoparticle Arrays with Strong Surface Lattice Resonance for Plasmonic Sensing. *Adv. Sci.* **2024**, *11* (12), No. e2306239.
- (35) Wang, D.; Bourgeois, M. R.; Lee, W. K.; Li, R.; Trivedi, D.; Knudson, M. P.; Wang, W.; Schatz, G. C.; Odom, T. W. Stretchable Nanolasing from Hybrid Quadrupole Plasmons. *Nano Lett.* **2018**, *18* (7), 4549–4555.
- (36) Deng, S.; Zhang, B.; Choo, P.; Smeets, P. J. M.; Odom, T. W. Plasmonic Photoelectrocatalysis in Copper-Platinum Core-Shell Nanoparticle Lattices. *Nano Lett.* **2021**, *21* (3), 1523–1529.
- (37) Steiner, A. M.; Mayer, M.; Seuss, M.; Nikolov, S.; Harris, K. D.; Alexeev, A.; Kuttner, C.; König, T. A.; Fery, A. J. A. n. Macroscopic strain-induced transition from quasi-infinite gold nanoparticle chains to defined plasmonic oligomers. *ACS Nano* **2017**, *11* (9), 8871–8880.
- (38) Conti, Y.; Passarelli, N.; Mendoza-Carreño, J.; Scarabelli, L.; Mihi, A. Colloidal Silver Nanoparticle Plasmonic Arrays for Versatile Lasing Architectures via Template-Assisted Self-Assembly. *Adv. Opt. Mater.* **2023**, *11* (23), 2300983.
- (39) Xu, L.; Liang, H.-W.; Yang, Y.; Yu, S.-H. J. C. r. Stability and reactivity: positive and negative aspects for nanoparticle processing. *Chem. Rev.* **2018**, *118* (7), 3209–3250.
- (40) Goudeli, E.; Pratsinis, S. E. J. A. J. Crystallinity dynamics of gold nanoparticles during sintering or coalescence. *AIChE J.* **2016**, *62* (2), 589–598.
- (41) Yang, F.; Chen, Q.; Wang, J.; Chang, J. J.; Dong, W.; Cao, W.; Ye, S.; Shi, L.; Nie, Z. J. A. n. Fabrication of Centimeter-Scale Plasmonic Nanoparticle Arrays with Ultranarrow Surface Lattice Resonances. *ACS Nano* **2023**, *17* (1), 725–734.
- (42) Zou, R.; Zhang, Q.; Zhao, Q.; Peng, F.; Wang, H.; Yu, H.; Yang, J. J. C.; Physicochemical, S. A.; Aspects, E. Thermal stability of gold nanorods in an aqueous solution. *Colloids Surf., A* **2010**, *372* (1–3), 177–181.
- (43) Offermans, P.; Schaafsma, M. C.; Rodriguez, S. R.; Zhang, Y.; Crego-Calama, M.; Brongersma, S. H.; Rivas, J. J. A. n. G. Universal scaling of the figure of merit of plasmonic sensors. *ACS Nano* **2011**, *5* (6), 5151–5157.
- (44) Reshef, O.; Saad-Bin-Alam, M.; Huttunen, M. J.; Carlow, G.; Sullivan, B. T.; Ménard, J.-M.; Dolgaleva, K.; Boyd, R. W. J. N. I. Multiresonant high-Q plasmonic metasurfaces. *Nano Lett.* **2019**, *19* (9), 6429–6434.
- (45) Huttunen, M. J.; Dolgaleva, K.; Törmä, P.; Boyd, R. W. J. O. e. Ultra-strong polarization dependence of surface lattice resonances with out-of-plane plasmon oscillations. *Opt. Express* **2016**, *24* (25), 28279–28289.
- (46) Humphrey, A. D.; Barnes, W. L. J. P. R. B. Plasmonic surface lattice resonances on arrays of different lattice symmetry. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90* (7), 075404.
- (47) Mayer, M.; Schnepf, M. J.; König, T. A. F.; Fery, A. J. A. O. M. Colloidal Self-Assembly Concepts for Plasmonic Metasurfaces. *Adv. Opt. Mater.* **2019**, *7* (1), 1800564.
- (48) Gupta, V.; Probst, P. T.; Gößler, F. R.; Steiner, A. M.; Schubert, J.; Brasse, Y.; König, T. A.; Fery, A. J. A. a. m. interfaces, Mechanotunable surface lattice resonances in the visible optical range by soft lithography templates and directed self-assembly. *ACS Appl. Mater. Interfaces* **2019**, *11* (31), 28189–28196.
- (49) Chu, Y.; Schonbrun, E.; Yang, T.; Crozier, K. B. Experimental observation of narrow surface plasmon resonances in gold nanoparticle arrays. *Appl. Phys. Lett.* **2008**, *93* (18), 181108.
- (50) Zhou, W.; Odom, T. W. Tunable subradiant lattice plasmons by out-of-plane dipolar interactions. *Nat. Nanotechnol.* **2011**, *6* (7), 423.
- (51) Wang, D.; Bourgeois, M. R.; Guan, J.; Fumani, A. K.; Schatz, G. C.; Odom, T. W. Lasing from Finite Plasmonic Nanoparticle Lattices. *ACS Photonics* **2020**, *7* (3), 630–636.
- (52) Lin, Y.; Wang, D.; Hu, J.; Liu, J.; Wang, W.; Guan, J.; Schaller, R. D.; Odom, T. W. Engineering Symmetry-Breaking Nanocrescent Arrays for Nanolasing. *Adv. Funct. Mater.* **2019**, *29* (42), 1904157.
- (53) Kravets, V. G.; Schedin, F.; Kabashin, A. V.; Grigorenko, A. N. Sensitivity of collective plasmon modes of gold nanoresonators to local environment. *Opt. Lett.* **2010**, *35* (7), 956–958.
- (54) Qi, X.; Pérez, L. A.; Alonso, M. I.; Mihi, A. High Q-Factor Plasmonic Surface Lattice Resonances in Colloidal Nanoparticle Arrays. *ACS Appl. Mater. Interfaces* **2024**, *16* (1), 1259–1267.
- (55) Gerislioglu, B.; Dong, L.; Ahmadvand, A.; Hu, H.; Nordlander, P.; Halas, N. J. Monolithic Metal Dimer-on-Film Structure: New Plasmonic Properties Introduced by the Underlying Metal. *Nano Lett.* **2020**, *20* (3), 2087–2093.